

LOVELY PROFESSIONAL UNIVERSITY

Group No.: 7

Title: Network Design for ABC Company

Subject: Network Administration

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Members:

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NETWORK DESIGN FOR ABC COMPANY

1. OBJECTIVE:

The objective of this assignment is to design a network for Company ABC that meets all requirements while considering media distance limitations, network device selection, environmental factors, and compatibility with existing and future equipment.

Case Study Scenario:

Company ABC leases three buildings (building A, B and C) in a large office park. The office park has a conduit system that allows physical media to run between buildings. The distance of building A to B is 1 km, A to C is 0.5 km and B to C is 1.12 km. Company ABC will use the Class B address of 172.16.0.0/16 for its sites. You should subnet this classful network not only to support the three buildings (one subnet per floor building), but to allow as many as six total sites in the future, as Company ABC continues to grow.

2. STEPS FOR SUBNETTING:

Step 1:
Determine the number of required subnets

- Building A: 3 floors = 3 subnets.
- Building B: 2 floors = 2 subnets.
- Building C: 1 floor = 1 subnet.

So in total we need 6 subnets for the current setup. But since the company plan to grow, we will plan for 6 sites, and each site has a similar setups of 3 fllors, so maybe potentially 18 subnets in total

Step 2: Number of hosts per Subnet

Building	Hosts per floor	No. of Floors
A	150, 100, 50	3
B	100, 100	2
C	100	1

Step 3: Subnet Calculation:

- Building A:
  - Floor 1:
    Number of needed host: 150 => 2^N >= 150 is 2^8 = 256 with 8 host bits

And the => Subnet bits will be DH - H = 16 – 8 = 8
And CIDR will be Subnet Bits + DN = 16 + 8 => /24 (255.255.255.0)
So for Floor 1 we have only 254 usable IP addresses

- Floor 2:
  Number of needed host: 100 => 2^N >= 100 is 2^7 = 128 with 7 host bits
  And the => Subnet bits will be DH - H = 16 – 7 = 9
  And CIDR will be Subnet Bits + DN = 16 + 9 => /225 (255.255.255.128)
  So for Floor two we will have 124 usable IP address

NOTE: ***The above calculation we be the same for Building B, both Floors and Building C floor 1 as they have the same number of hosts***

- Floor 3:
  Number of needed host: 50 => 2^N >= 150 is 2^6 = 64 with 6 host bits
  And the => Subnet bits will be DH - H = 16 – 6 = 10
  And CIDR will be Subnet Bits + DN = 16 + 10 => /225 (255.255.255.192)
  So for Floor two we will have 62 usable IP address.

Step 4: Assigning Network Addresses:

Starting with the base network address 172.16.0.0/16, let's assign subnets for each floor:

- Building A:
  - Floor 1 (150 hosts): 172.16.0.0/24
    Usable IP range: 172.16.0.1 to 172.16.0.254
  - Floor 2 (100 hosts): 172.16.1.0/25

    Usable IP range: 172.16.1.1 to 172.16.1.126
  - Floor 3 (50 hosts): 172.16.1.128/26

    Usable IP range: 172.16.1.129 to 172.16.1.190
- Building B:
  - Floor 1 (100 hosts): 172.16.2.0/25

    Usable IP range: 172.16.2.1 to 172.16.2.126
  - Floor 2 (100 hosts): 172.16.2.128/25

    Usable IP range: 172.16.2.129 to 172.16.2.254
- Building C:
  - Floor 1 (100 hosts): 172.16.3.0/25

    Usable IP range: 172.16.3.1 to 172.16.3.126

Addressing table:

Building	Floor	Hosts	Network Address	Subnet Mask	Usable IP Range
A	Floor 1	150	172.16.0.0	/24(255.255.255.0)	172.16.0.1 to 172.16.0.254
A	Floor 2	100	172.16.1.0	/25(255.255.255.128)	172.16.1.1 to 172.16.1.126
A	Floor 3	50	172.16.1.128	/26(255.255.255.192)	172.16.1.129 to 172.16.1.190
B	Floor 1	100	172.16.2.0	/25(255.255.255.128)	172.16.2.1 to 172.16.2.126
B	Floor 2	100	172.16.2.128	/25(255.255.255.128)	172.16.2.129 to 172.16.2.254
C	Floor 1	100	172.16.3.0	/25(255.255.255.128)	172.16.3.1 to 172.16.3.126

Future Subnets will start from 172.16.4.0/24 upward.

3. DEVICES USING IN DIFFERENT LAYERS

Layer 1 (Physical Layer) Media Selection:

Fiber Optic Cable: Given the distances between buildings (1 km, 0.5 km, and 1.12 km), fibe r optic cables are ideal. They support high data rates over long distances and are less suscepti ble to electromagnetic interference.

For inter-building connections (A-B, A-C, B-C) due to the distances involved.

Layer 2 (Data Link Layer) Device Selection:

Switches:

- **Cisco Catalyst 2960 Series Switches:** These are reliable and cost-effective, supporting the needs of each building. Each floor will have its own switch t o handle the local network traffic and connect the wireless access points.
- Cisco Catalyst 2960-X Series Switches: These switches will be placed on each floor of buildings A, B, and C to handle local network traffic and connect wireless access points

Layer 3 (Network Layer) Device Selection:

Routers:

- **Cisco ISR 4000 Series Router:** This router can handle the internet connection (at lea st 30Mbps) and manage the inter-building communication efficiently.
- Cisco ISR 4331 Router: This router will be installed in building A to handle the internet connection and manage inter-building communication

4. WIRELESS DEVICES :

- **Wireless Access Points (WAPs):** For Wi-Fi coverage in each floor of the buildings, using something like the **Cisco Aironet 18 50 Series Access Points** can provide robust and scalable wireless connectivity.
- Cisco Aironet 1852i Access Points: These access points will provide robust and scalable wireless connectivity for each floor

Each building contains various Wi-Fi client devices for example, smart-phones, tablets, and laptops

5. ENVIRONMENTAL CONSIDERATIONS:

- **Climate Control**: Ensure switches and routers are placed in climate-controlled environments to avoid overheating.
- **Cable Management**: Use proper conduits and cable management systems to protect p hysical media from damage.
- **Power Backup**: Install Uninterruptible Power Supplies (UPS) to keep critical networ k infrastructure operational during power outages.

Compatibility:

- **Current Equipment**: Ensure all new devices are compatible with existing devices in use.
- **Future Proofing**: Choose devices that support higher data rates and more users to acc ommodate future growth.

6. DESIGN

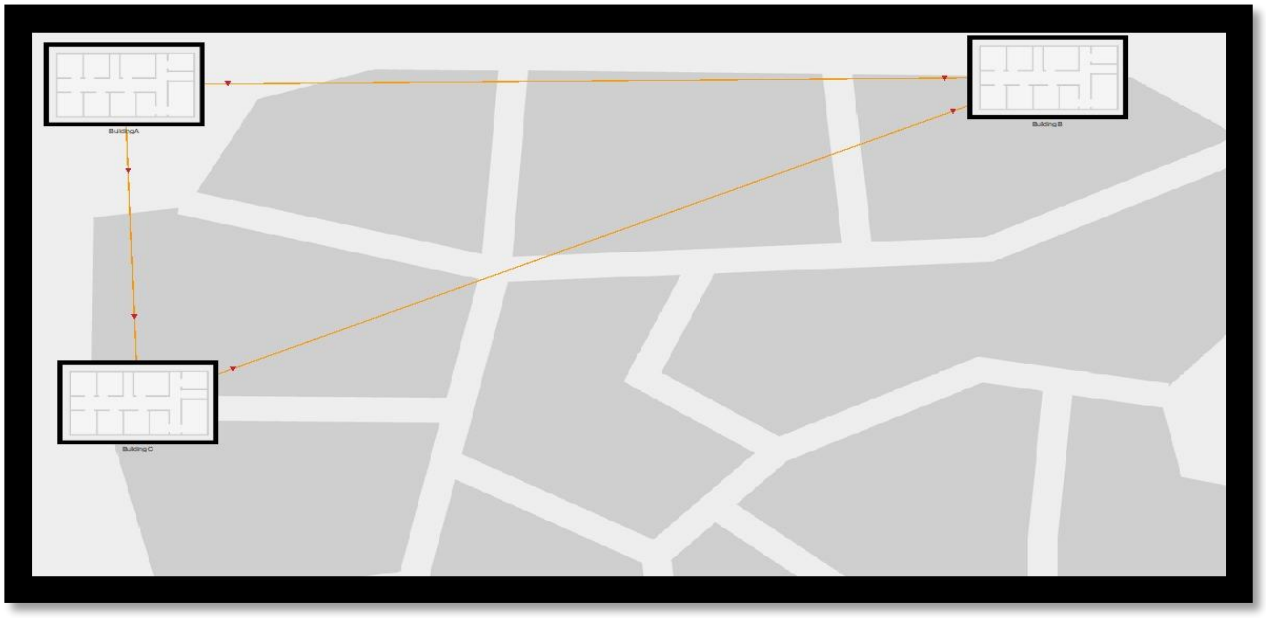


Fig 1: Design for 3 buildings

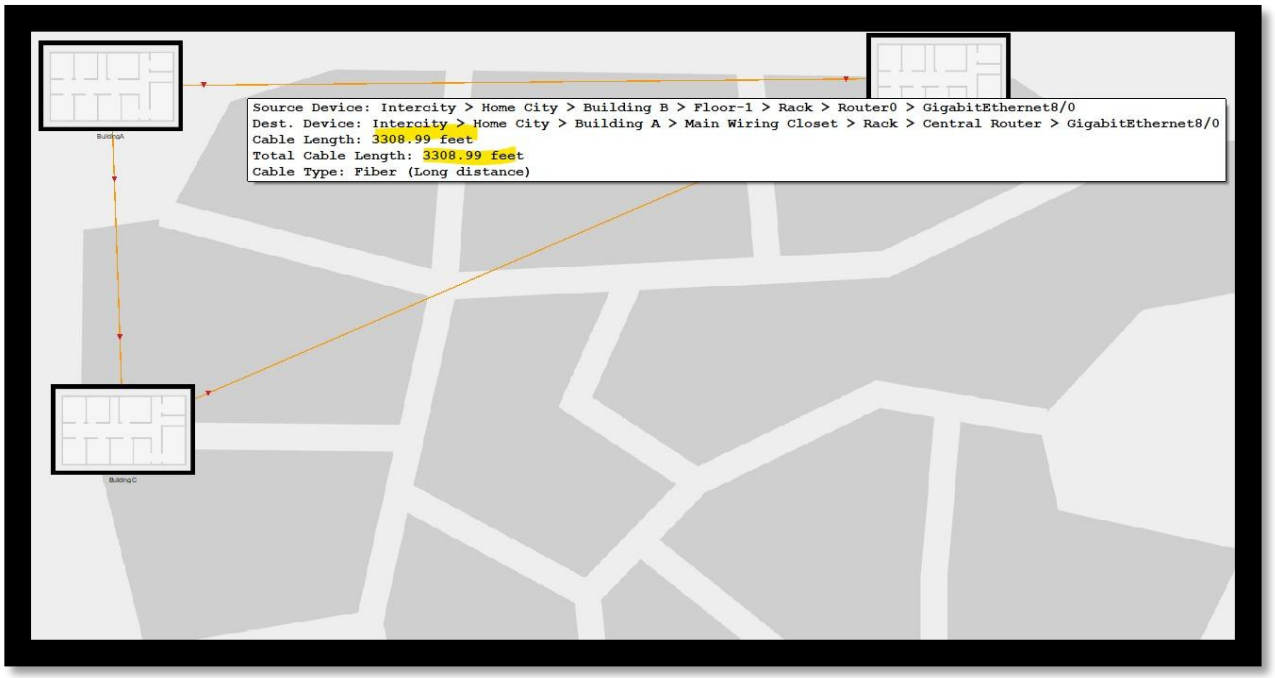


Fig 2; Show the distance between Building A and Building B

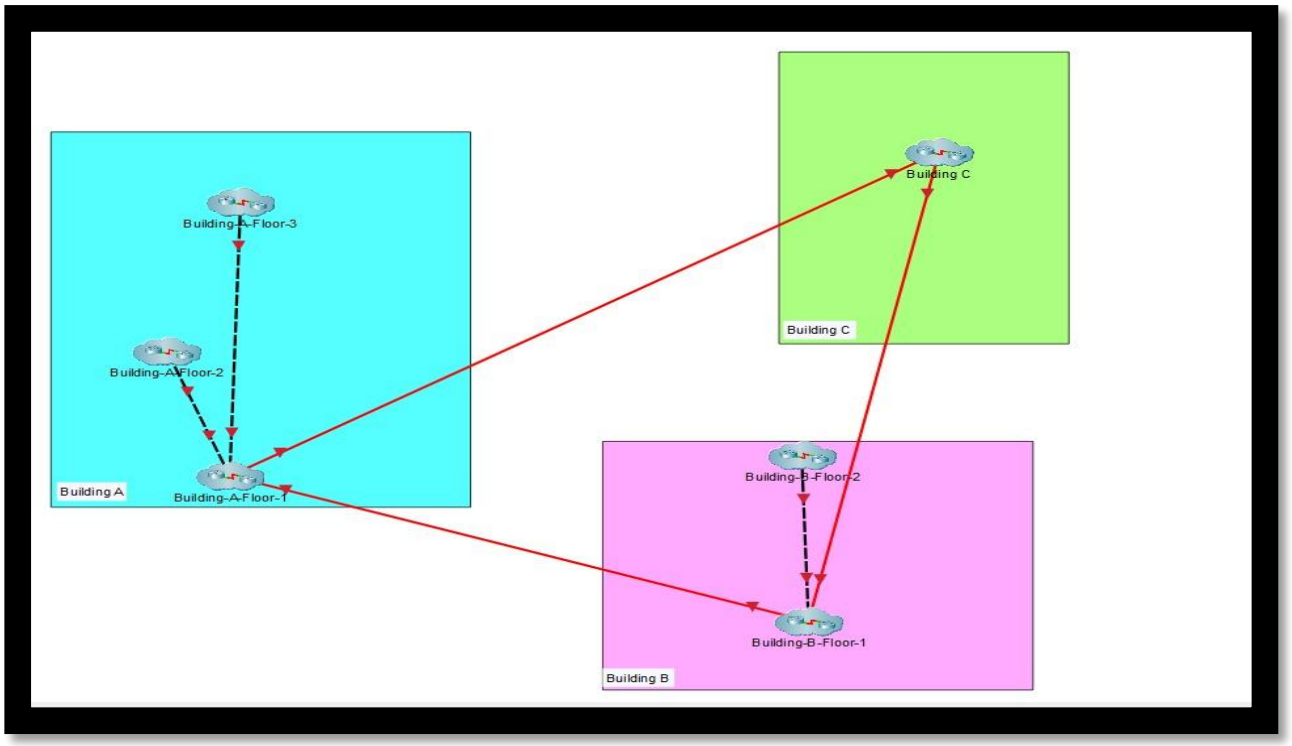


Fig 3; Logical Design

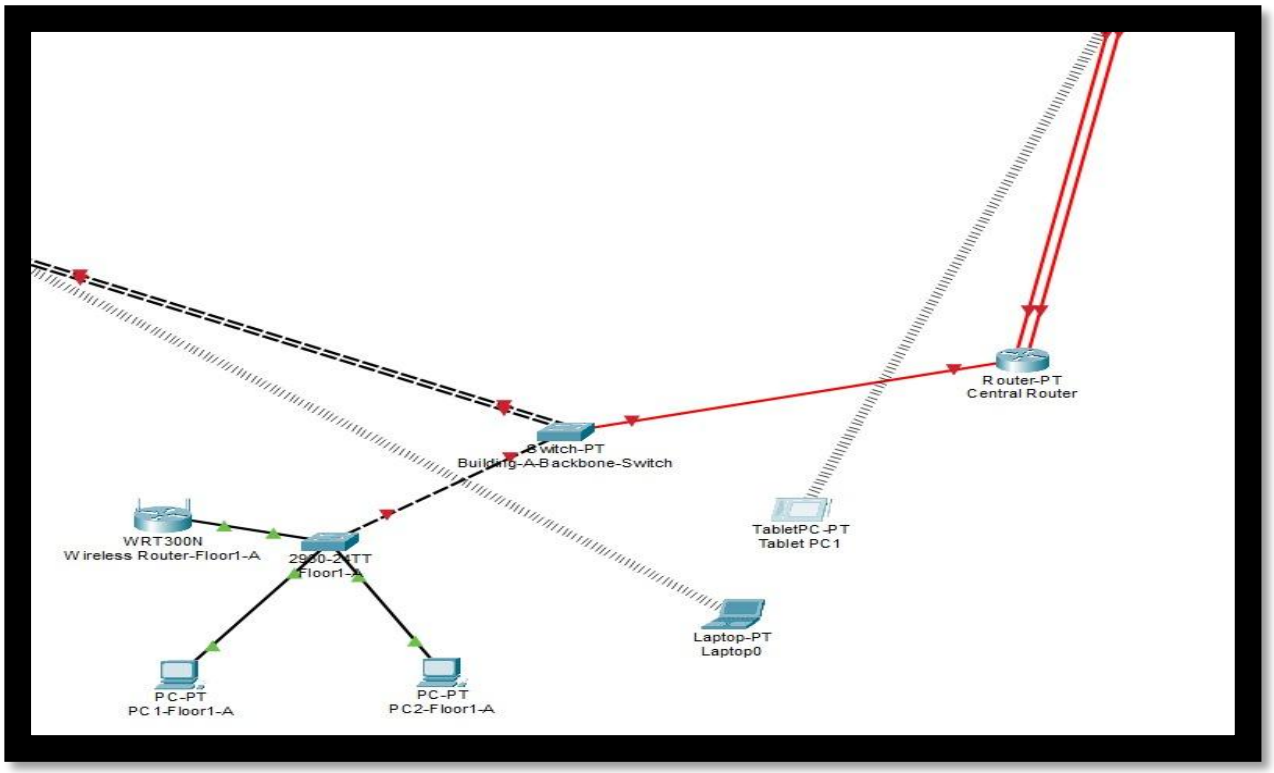


Fig 4; Inter connections